

PHL 1010 Tutorial III

07/09/2025 (Friday)

Note: For problem 3, 4 and 5, you are not supposed to use Gauss' Law.

1. Construct a vector field that has nonzero divergence as well as nonzero curl everywhere. (well, you may get hint from the class notes! Construct a field which is more interesting and not discussed in the class)
2. A vector field is given by $\vec{V} = (x^2 - yz + y)\hat{x} + (x - 2yz)\hat{y} + (z^2 - 2zx + x + y)\hat{z}$. Is this field irrotational or solenoidal? Find a scalar/vector potential as the case may be.
3. (a) Calculate the electric field at a point P (0, 0, h) due to an infinite sheet of charge along the $z = 0$ plane having a uniform charge density σ . (Hint: use cylindrical polar coordinate system)
(b) A circular disk of radius R is now removed from the above sheet. What would be the fractional change in the field intensity at the same location? Assume point P on the axis of the hole.
(c) Draw the line of force in 2D considering the side view of this charge distribution.
4. The finite sheet $0 \leq x \leq 1$, $0 \leq y \leq 1$ on the $z = 0$ plane has a charge density $\sigma = xy(x^2 + y^2 + 25)^{3/2} \text{ nC/m}^2$. Calculate:
(a) the total charge on the sheet
(b) electric field at (0, 0, 5)
(c) force experience by a -4mC charge located at (0, 0, 5).
5. A sphere of radius "a", carrying a uniform volume charge density ρ is placed in a region with its center at the origin. Find the electric field at a point (0, 0, z) from the center of this sphere. Express your answer in terms of the total charge of the sphere "q". Draw the graph of $|\vec{E}|$ as a function of the distance from the center. (Hint: use the cosine law wherever needed)
6. Evaluate the following integral:
(a) $\int_V |\vec{r} - \vec{a}|^2 \delta(2\vec{r}) d\tau$ over a unit cube centered at the origin. Further, $\vec{a} = 2\hat{x} - \hat{y}$.
(b) $\int_V e^{-r} \left(\vec{\nabla} \cdot \frac{\hat{r}}{r^2} \right) d\tau$, where V is the volume of sphere of radius R centered at origin.

Home Exercise: 2.10, 2.14, 2.15

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